Open-Source Reward Distillation for Semantic Exploration in SENSEI

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Abstract

SENSEI is a Reinforcement Learning framework that uses VLM annotations to distill a reward function with Motif. This reward function is used to guide the semantic exploration of a DreamerV3 agent. However, one shortcoming of this framework is that it uses the close-sourced and usage-cost inducing GPT-4. This limits the accessibility and scalability of semantic exploration in SENSEI. This study investigates whether LLaVA, an open-source VLM, can replace GPT-4 within this framework. We deploy SENSEI in the Robodesk environment to analyze how the resulting reward model from LLaVA annotations compares to the GPT-4 counterpart. We find that LlaVA annotations can guide exploration, but are not as effective as GPT-4 annotations.

1 Introduction

Exploration is a key challenge in reinforcement learning (RL), especially when dealing with environments with sparse or delayed rewards (16; Hao et al.). Using intrinsic motivation, agents can explore an environment without relying on external rewards. The recent RL framework Semantically Sensible Exploration (SENSEI) incorporates this motivation by using vision-language models (VLMs) to provide semantic feedback (12). It uses GPT-4 to annotate image pairs that show the agent interacting with its environment, stating which image is more interesting. The framework then distills an intrinsic reward function from these annotations, which can optimize exploration. The limitation of SENSEI is that GPT-4 is closed-source and costly. In this project, we explore whether the Large Language and Vision Assistant (LLaVA), an open-source VLM by Liu et al. (8), could serve as a solid alternative to GPT-4. This replacement can improve accessibility and scalability to semantic exploration.

2 Background & Related Work

2.1 Intrinsic Motivation in Reinforcement Learning

Intrinsic motivation techniques allow RL agents to explore their environment without external rewards. Three classic approaches to intrinsic motivation are state-space coverage (2; 15; 3), novelty-driven methods (9; 13), and information-gain strategies (10; 14; 11). In state-space coverage, agents are rewarded for visiting rarely seen states, whereas novelty-driven methods such as random network distillation (RND) reward agents based on prediction errors against learned or random networks. Information-gain, on the other hand, reduces uncertainty about environment dynamics. These methods often lack semantic understanding of what makes exploration meaningful, and usually only discover low-level behaviors, which might not align with downstream tasks (12). Recent work has focused on injecting human priors via large language models (LLMs) in order to guide agents toward behaviors meaningful to humans using textual knowledge (4). VLMs can overcome the need for preexisting semantic labels and high-level actions that such LLM's have, by extracting task-relevant signals directly from visual observations (1).

2.2 SENSEI Framework

Semantic exploration in SENSEI is achieved in three phases: (1) image annotation, (2) reward distillation, and (3) model-based exploration. Image annotation uses GPT-4 to annotate pairs of observations from an unlabeled dataset by prompting "Which of the two images is more interesting?". The exact prompt is available in A. These pairwise judgments are then distilled into a scalar reward function R_ψ by minimizing the cross-entropy loss using Motif (7). During training, the semantic reward is combined with an uncertainty-based reward (14) and is given to the model-based agent. The agent is a recurrent state-space model (RSSM); which is built directly on the Dreamer V3 framework (5). SENSEI's intrinsic reward is a mix of the semantic signal r_{sem} and the uncertainty bonus r_{dist}. When the semantic reward is low the agent prioritizes seeking semantically rich states and, once that reward exceeds a predefined threshold, it shifts focus to maximizing uncertainty to branch into novel regions. The intrinsic reward is balanced using the formula present in B.

2.3 LLaVA and Open-Source VLMs

This study uses the open-source VLM LLaVA, which combines a frozen CLIP ViT-L/14 encoder with the Vicuna LLM through a lightweight projection matrix (8). Several papers have explored using open-source VLMs in interactive agents. For example, OmniJARVIS unifies vision–language–action tokens in an autoregressive transformer to reason, plan, and act in the open-world videogame Minecraft (17).

3 Methodology & Experiments

This study works with the Robodesk environment, which consists of a tabletop with various objects, such as drawers and buttons. It provides rich visual observations and low-level control using joint actions, which makes it ideal for evaluating semantic exploration of SENSEI. LLaVA-v1.5-7b with 7.3 billion parameters is used to replace gpt-4-turbo-2024-04-09 with 1.8 trillion parameters. Around 100 thousand image pairs from previous Robodesk episodes are passed to LLaVA, which labels image pairs with preferences. Next, the image pairs and their labels are passed to Motif which uses them to train a reward model. The model checkpoints are stored for later integration with DreamerV3. This integration completes the SENSEI framework. The separate configurations for LLaVA, Motif and DreamerV3 are available for reference in C. The Motif and DreamerV3 configurations are the same as in the original paper.

Our approach is first evaluated by comparing the annotations made by LLaVA and GPT-4. The quality of the LLaVA annotated reward function is then evaluated using task-free exploration in the Robodesk environment. As baselines, all evaluated methods from the original paper are used. In addition, we reproduce the results of the original SENSEI from the paper by using an identical GPT-4 distilled reward function and DreamerV3 configuration. Semantically meaningful behavior is assessed in terms of interaction counts with all separate objects.

4 Results & Discussion

The LLaVA image-pair annotations display a clear bias for the first image, which is chosen around 80% of the time. GPT-4, on the other hand, is slightly biased towards the second image, which is preferred in 60% of the cases. The exact numbers can be seen in D. A possible reason for this bias could be that LLaVA has far fewer parameters than GPT-4.

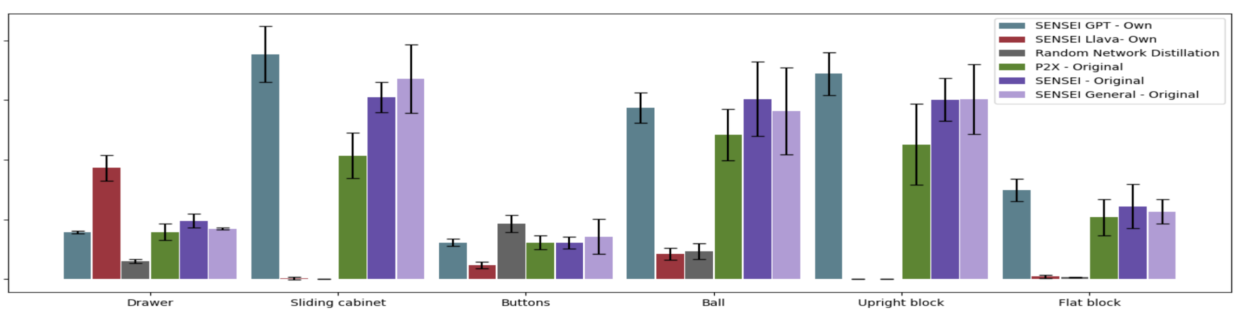

Figure 1: Mean number of interactions during 250K steps for various methods together with standard deviation across 3 seeds. SENSEI General is not given a description of the environment, instead it is prompted to generate one themselves.

All relevant Robodesk results are shown in Figure 1. When using GPT-4 annotations, the obtained results resemble those from the original paper, with minor seed-related variance.

When using LLaVA annotations, interactions with specific objects change significantly compared to the GPT-4 baseline. For example, LLaVA annotations lead to about twice as many interactions with the drawer as the baseline, which suggests that LLaVA is able to distill the reward function with semantically meaningful behavior. The larger amount of interactions for the drawer can be explained by "drawer" being explicitly mentioned in the prompt, which does not hold for all objects. The smaller parameter count of LLaVA might prevent it from generalizing past this explicit mention. Apart from the drawer, the interaction counts of other objects are significantly lower than that of the GPT-4 baseline. This indicates that LLaVA either has a strong preference for the drawer, lacks interest in other objects than the drawer, or exhibits both tendencies.

5 Conclusion & Future Work

This study demonstrated that LLaVA, a fully open-source VLM, is capable of distilling a notion of "interestingness" that can guide exploration in RL. Despite its smaller model size compared to GPT-4, LLaVA produces annotations that can influence the behavior of a DreamerV3 agent, particularly when the semantic information in the prompt is clearly stated. The observed bias towards the drawer interactions shows both the strengths and weaknesses of LLaVA: It is able to effectively guide the reward function toward semantically meaningful behavior, but struggles to generalize beyond the behavior explicitly mentioned in the prompt. This annotation bias is likely caused by the smaller parameter count.

Nonetheless, LLaVA represents a viable alternative for semantic exploration when in need of an open-source and cost-efficient VLM. The successful integration of LLaVA into the SENSEI framework opens new opportunities to incorporate intrinsic motivation into RL. By removing the dependency on closed-source VLMs such as GPT-4, SENSEI becomes more accessible, democratized, and scalable for further research and real-world applications.

5.1 Future Work

Future work could focus on using newer and more powerful open-source VLMs such as LLaVa-Next, refining the prompt design by experimenting with explicit mentions of interesting objects, using other environments such as Atari from OpenAI Gym, and experimenting with hyperparameter settings to optimize reward function distillation. Such research can possibly reduce the annotation bias evident from this study and enhance semantic exploration, thus narrowing the performance gap between open-sourced and closed-sourced SENSEI variants.

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A Prompt

"Consider these two images.
1. What is shown in the top (upper) image? Is the robot gripper very close to objects on the desk or the drawer?
2. What is shown on the bottom (lower) image? Is the robot gripper close to objects on the desk or the drawer?
3. The goal is for the robot to explore interesting things and interact with objects and drawers.
Is there any difference between the top and bottom image in terms of how interesting it is?
Think step by step and answer based on what you see.
Remember: If the robot arm is retracted or away from the desk, there is no interaction and it is not interesting.
The closer the robot gripper is to objects or the drawer, the more interesting the image is. Hovering above the desk is not more interesting than actively reaching out to a certain object."

B Intrinsic Reward

$$r_t^{\text{expl}} = \begin{cases} r_t^{\text{sem}} + \beta_{\text{go}}\, r_t^{\text{dist}}, & r_t^{\text{sem}} < Q_k, \\ r_t^{\text{sem}} + \beta_{\text{explore}}\, r_t^{\text{dist}}, & r_t^{\text{sem}} \geq Q_k, \end{cases} \tag{1}$$

Equation 1: Adaptive Go–Explore intrinsic reward. The agent blends a semantic "interestingness" bonus r_t^{sem} with an uncertainty bonus r_t^{dist}, using coefficients β_{go} and β_{explore}. A running kth-quantile threshold Q_k on the semantic reward determines whether the policy prioritizes reaching semantically meaningful states ("go" phase) or exploring areas of high uncertainty ("explore" phase).

C Configurations

Config 1: Configuration file for annotating image pairs with LLaVA

```
annotator: "llava"
annotator_params:
    model_params:
        model_path: "liuhaotian/llava-v1.5-7b"
        load_in_8bit: false
        save_img: true
        image_concat_orient: "vertical"
        image_concat_buffer: 28
    sampling_params:
        temperature: 0.6
        top_p: 0.8
        max_new_tokens: 1024
dataset_dir: "/home/scur2659/motif_group24/robodesk_data"
partition_size: 10000
partition_idx: 0
logging: true
seed: 42
device: "cuda:0"
working_dir: "/home/scur2659/motif_group24/robodesk_data/llava_annotations"
```

Config 2: Configuration file for distilling reward function with Motif (Same for GPT-4 and LLaVA experiments)

```
reward_model_params:
    model_params:
        encoder_params:
            encoder_num_layers: 2
            encoder_hidden_dim: 512
            use_obs_vec: false
            use_clip_embedding_right: false
            use_clip_embedding_left: false
```

```
            use_image: true
            resize_image: false
            normalize_inputs: false
            image_filter_max: 64
        param_initialization: "orthogonal"
        param_init_gain: 1
    train_params:
        weight_decay: 0
        learning_rate: 0.00003
        batch_size: 32
        num_epochs: 50
dataset_dir: "robodesk_data"
preference_key: "preferences_llava"
seed: 42
working_dir: "results/reward_model_with_llava_anotation"
device: "cuda:0"
```

Config 3: Configuration file for deploying SENSEI in Robodesk (Same for GPT-4 and LLaVA experiments)

```
robodesk_sensei:
  run:
    script: train_eval
    eval_every: 5e4
    steps: 250000
    report_every: 5e4
    log_every: 5e3
    log_keys_sum: '^rew_.*'
    log_keys_mean: 'motif_reward'
    log_zeros: True
    expl_until: 2e6
  grad_heads: [decoder, cont]
  task: motifrobodesk_open_drawer_medium_sparse
  env.motifrobodesk:
    motif_model_dir: '/motif/trained_motif_networks/robodesk_intersection_p2e_gpt'
    model_cpt_id: "15"
  expl_rewards:
    extr: 0.0
    disag: 1.0
    vlm_reward: 0.1
  perc_based_scaling:
    perc_scale: True
    perc_key: "vlm_reward"
    perc_threshold: 75.0
    alt_scales: { disag: 0.0, vlm_reward: 0.1, extr: 0.0 }
  vlm_reward_key: 'motif_reward'
  add_reward_heads:["motif_reward", "rew_open_slide", "rew_open_slide_easy",
  "rew_open_drawer", "rew_open_drawer_medium", "rew_open_drawer_easy",
   "rew_push_green", "rew_stack", "rew_upright_block_off_table", "rew_flat_block_in_bin",
   "rew_flat_block_in_shelf", "rew_lift_upright_block", "rew_lift_ball"]
  run.train_ratio: 512
  rssm.deter: 1024
  .*\.cnn_depth: 48
  .*\.units: 640
  .*\.layers: 3
  encoder: {mlp_keys: '$^', cnn_keys: 'image'}
  decoder: {mlp_keys: '$^', cnn_keys: 'image'}
```

D VLM preference distribution

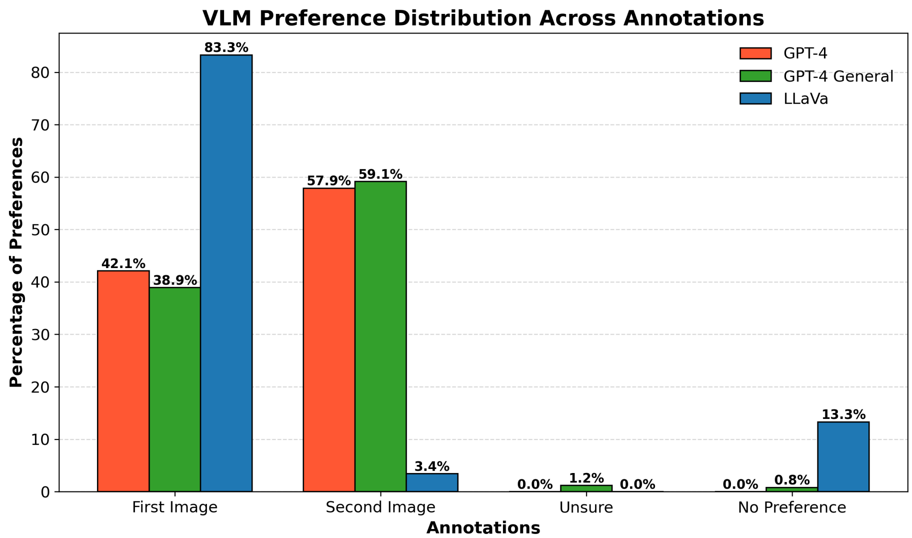

Figure 2: Bar Chart showing the Preference Distribution in Percentages across different VLMs (GPT-4, GPT-4 General, and LLaVa) for Four Types Annotations: First Image, Second Image, Unsure, or No Preference.